Bharatiya Vidya Bhavan

BHAVAN'S VIVEKANANDA COLLEGE
OF SCIENCE, HUMANITIES AND COMMERCE
(Accredited with 'A' Grade by NAAC)
Autonomous College – Affiliated to Osmania University
Department of Computer Science

PROGRAM NAME: B.Sc. (Computer Science) (w.e.f 2022-23)
COURSE NAME: Programming in Java

PAPER CODE: CS525 **PPW: 4**
YEAR/SEMESTER: III/V **NO. OF CREDITS: 4**

COURSE OBJECTIVE: To enable students with the concepts of Java Programming and develop GUI applications.

UNIT-WISE COURSE OBJECTIVES:

COb1: To discuss the features of Java and construct class programs with methods.
COb2: To illustrate types of Inheritance, Packages and Arrays concepts.
COb3: To explore the concepts of Exception handling, Multithreading and Input/Output.
COb4: To learn the concepts of Applets, AWT and Swings.

UNIT-I: **15 hrs**
Introduction: Java Essentials, JVM, Java Features, Creation and Execution of Programs, Data Types, Structure of Java Program, Type Casting, Classes, Objects, Class Declaration, Creating Objects.
Method Declaration and Invocation, Method Overloading, Constructors – Parameterized Constructors, Constructor Overloading, Cleaning-up unused Objects.
Class Variables &Method-static Keyword, this Keyword and Command-Line Arguments.
(Ch: 2.4, 2.5, 2.6, 2.7, 3.2, 3.8, 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, 4.11)

UNIT-II: **15 hrs**
Inheritance: Introduction, Types of Inheritance, extends Keyword, Examples, Method Overriding, super, final Keyword, Abstract classes, Interfaces, Abstract Classes Verses Interfaces.
Packages: Creating and Using Packages, Access Protection.
Arrays: One-Dimensional Arrays, Two-Dimensional Arrays, Wrapper Classes, String Class.
(Ch: 5.1.1, 5.1.2,5.2,5.3,5.4,5.5,6.1,6.1.3,6.2,4.10,6.3.2,6.3.3)

UNIT-III: **15 hrs**
Exception: Introduction, Types, Exception Handling Techniques-try, catch, multiple catch, User-Defined Exception.
Multithreading: Introduction, Main Thread and Creation of New Threads –By Inheriting the Thread Class, Thread Lifecycle, Thread Priority.

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BOS in Computer Science
Bhavan's Vivekananda College
Sainikpuri

Dr. N. KISHAN
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...SITY
HYDERABAD-500 007. T.S.

Input/Output: Introduction, java.io Package, Reading and Writing Data- Reading/Writing Console User Input, Scanner Class, Reading/Writing Buffered Byte Stream Classes-Buffered Input Stream Class, Buffered Output Stream Class.
(Ch: 7.1, 7.2, 7.3, 8.1, 8.4, 8.5, 8.6, 8.7, 9.1, 9.2, 9.3.2. 9.3.4)

UNIT-IV: **15 hrs**

Applets: Introduction, Example, Life Cycle, Applet Class, Common Methods Used in Displaying the Output (Graphics Class).
AWT: Introduction, Components, Containers, Button, Label, Checkbox, Radio Buttons, Text Field and Text Area.
Event Handling: Introduction, Event Delegation Model, Events Classes- Action Event, Key Event, Mouse Event, Mouse Wheel Event. Event Listeners-ActionListener, KeyListener, MouseListener, MouseWheelListener.
Swings: Introduction, Differences between Swing and AWT, JFrame, JApplet, JPanel.
(Ch: 12.1, 12.4, 12.5, 12.2, 12.6, 14.1, 14.2, 14.3, 14.4, 14.5, 14.6, 14.9, 13.1, 13.2, 13.3.1,13.5,15.1,15.1.2, 15.2, 15.3, 15.4)

PRESCRIBED BOOK:

1. **Programming in Java**, Sachin Malhotra, Saurabh Choudhary, Oxford University Press, Second edition, 2018.

REFERENCE BOOKS:

2. **Thinking in Java**, Bruce Eckel, Pearson Edition, Fourth Edition, 2008.
3. **Java: The Complete Reference**, Herbert Scheldt , Tata McGraw Hill; Eleventh edition,2020.
4. **Introduction to Java Programming**, Y. Daniel Liang, Pearson Education; Tenth edition, 2018.
5. **Java: How To Program**, Paul Deitel, Harvey Deitel, Pearson Education; Eleventh edition , 2018.
6. **Core Java Volume I –Fundamentals**,Cay S. Horsttnann, Pearson Education; 11th edition, 2020.

COURSE OUTCOMES:

At the end of the course students will be able to:

CO1: Comprehend the features of Java and construct class programs with methods.
CO2: Apply the concepts of Inheritance, Packages and Arrays concepts.
CO3: Program the concepts of Exception handling, Multithreading and Input/Output.
CO4: Develop GUI programs using Applets, AWT and Swings.

CHAIRPERSON
BOS in Computer Science
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Dr. N. KISHAN
Professor & Head
Department of Mathematics
OSMANIA UNIVERSITY
HYDERABAD-500 007, T.S.

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PROGRAM NAME: B.Sc. (Computer Science) (w.e.f 2022-23)
COURSE NAME: Programming in Java Lab

PAPER CODE: CS525P **PPW: 4**
YEAR/SEMESTER: III/V **NO. OF CREDITS: 1**

COURSE OBJECTIVE: To enable students to apply Object-Oriented Concepts and develop GUI applications.

COb1: Learn to program concepts of OOPs, Arrays, Exception handling
COb2: To illustrate the concepts of Multithreading, Input/Output, Applets, AWT and Swings

Note:

- Programs of all the Concepts from Text Book including exercises must be practiced and executed.
- In the external lab examination student has to execute two programs with compilation and deployment steps are necessary.
- External Vice-Voce is compulsory.

Week 1:

1. Write a program to find whether a given number is prime or not.
2. Write a menu driven program for following:
   a) Display a Fibonacci series
   b) Compute Factorial of a number

Week 2:

3. Write a program to create an array of 10 integers. Accept values from thè user in that Array. Input another number from the user and find out how many numbers are equal to the number passed, how many are greater and how many are less than the number passed.
4. Write java program for the following matrix operations:
a) Addition of two matrices b) Transpose of a matrix

Week 3:

5. Write a java program that computes the area of a circle, rectangle and a Cylinder using Method overloading.
6. Write a program to demonstrate about types of constructors.

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Week 4:

7. Write a program to demonstrate about inner classes.
8. Write a program to demonstrate Method overriding.

Week 5:

9. Write a Java program for the implementation of multiple inheritance using Interface to calculate the area of a rectangle and triangle.
10. Write a program to create a package called Arithmetic that deals with arithmetic operations.

Week 6:

11. Write a program to demonstrate throws and finally keywords.
12. Write a program that reads two integer numbers for the variables a and b. If any other character except number (0-9) is entered then the error is caught by NumberFormatException object. After that ex.getMessage () prints the information about the error occurring causes.

Week 7:

13. Create a class called Fraction that can be used to represent the ratio of two integers. Include appropriate constructors and methods. If the denominator becomes zero, throw and handle an exception.

Week 8:

14. Write a program for the following string operations:
    a. Compare two strings b. concatenate two strings c. Compute length of a string
15. Write a program to demonstrate StringBuffer class Methods.

Week 9:

16. Write a program to demonstrate Multithreading using Runnable Interface.
17. Write a program to demonstrate Synchronization in Multithreading.

Week 10:

18. Write a program to demonstrate FileInputStream and FileOutputStream Class.
19. Write a program to demonstrate RandomAccessFile class.

Week 11:

20. Write a java program to create a frame window in an Applet. Display your name,address and qualification in the frame window.(container class)
21. Write a java program to draw a line between two coordinates in a window. (container class)

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Week 12:

22. Write a java program to display the following graphics in an applet window.
    a. Rectangles b. Circlesc. Ellipses d. Arcse. Polygons

Week 13:

23. Write a java program to demonstrate Components in Swings.
24. Write a java program to demonstrate JTable in Swings.

Week 14:

25. Assignment on AWT and Swing Components.

COURSE OUTCOMES:

At the end of the course students will be able to:

COb1: Apply OOPs Concepts, Arrays and Exception handling.
COb2: Implement Multithreading, Input/Output, Applets, AWT and Swings.

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